

Jaeuk Moon

Staff Engineer, AI Model Team · Samsung Research

pqsolutionax@gmail.com · jaeukmoon.github.io · github.com/jaeukmoon · Seoul, Republic of Korea

WORK EXPERIENCE

Staff Engineer — Samsung Research (AI Model Team)

Sep 2025 – Present

- LLM post-training research (SFT, RLHF) improving alignment and deployability of Samsung's internal foundation LLM.
- Developing an on-device LLM for wearable health monitoring, applying time-series forecasting and self-supervised learning to noisy sensor data.
- Robotics research on vision-language-action (VLA) models and world models, coupling action-conditioned future prediction with policy learning.
- Leading independent research on LLM-based influenza (ILI) forecasting with multimodal inputs and self-instructed fine-tuning; first-author paper published in IEEE JBHI (MIFlu, 2025), follow-up submitted to IEEE BIBM 2026.

Ph.D. Researcher & Lab Manager — Multimedia Information Lab, Korea University

Mar 2019 – Aug 2025

- 18 international journal publications (8 first-author SCIE, highest in the lab), 11 patent filings, multiple conference papers.
- Improved COVID-19 projection accuracy by 200% in collaboration with KISTI and the Korea Disease Control and Prevention Agency.
- Applied explainable AI to industrial energy optimization, saving over \$18,000 annually at a partner factory.
- Spearheaded a hierarchical multi-AI-agent system for B2B services such as Incheon International Airport.
- As lab manager, streamlined research workflows and mentored junior researchers to 15 awards and 10 SCIE-indexed papers.

EDUCATION

Ph.D. in Electrical and Computer Engineering — Korea University

Mar 2019 – Aug 2025

- GPA 4.11/4.5 · Advisor: Prof. Eenjun Hwang · Dissertation on LLM-based explainable multimodal forecasting.

B.E. in Electrical Engineering — Korea University

Mar 2015 – Feb 2019

SELECTED FIRST-AUTHOR PUBLICATIONS (8 SCIE total)

MIFlu: Large Language Model-Based Multimodal Influenza Forecasting Scheme — IEEE JBHI, 2025

RESEAT: Recurrent Self-Attention Network for Multi-Regional Influenza Forecasting — IEEE JBHI, 2023

Model-Agnostic Meta-Learning-Based Region-Adaptive Parameter Adjustment for Influenza Forecasting — J. King Saud Univ. – CIS, 2023

Anomaly Detection Using MAML-Based Variational Auto-Encoder for Facility Management — J. Building Engineering, 2023

Conditional Tabular GAN-Based Two-Stage Data Generation for Short-Term Load Forecasting — IEEE Access, 2020

SELECTED PROJECTS

- **Deep-learning energy forecasting for microgrids** — National Research Foundation of Korea (2019–2023).
- **Infectious disease response technology** — deep-learning epidemic models with KHIDI (2019–2022).
- **LLM solutions & hierarchical AI agents** — fine-tuning, RAG, hallucination self-evaluation for enterprise services (2024–2025).

SKILLS & LANGUAGES

ML/AI: LLM post-training (SFT · RLHF · LoRA/PEFT), time-series forecasting, meta-learning, GANs, explainable AI

Engineering: Python, PyTorch, TensorFlow, distributed training (DeepSpeed), vLLM/serving, evaluation harnesses

Languages: Korean (native), English (fluent — OPIc AL)